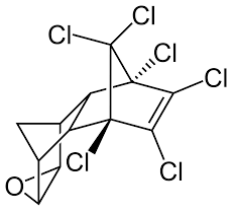
Analytical Chemistry Cumulative Exam (April 7, 2023)
Topic: HPLC and UPLC

1. (10 points)
(A) List at least two properties of chemical species that cause those species to be unsuitable for separation by Gas Chromatography (GC), but can still allow those species to be separated by High Performance Liquid Chromatography (HPLC), instead?

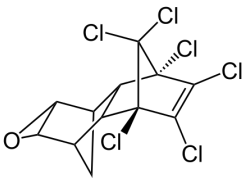
(B) Name three classes of compounds or name (or draw) three specific molecules within different classes of compounds that are examples of molecules that can be separated by HPLC but not by GC.

2. (30 points) Here are several types of liquid chromatography:
A = size exclusion,
B = partition chromatography (normal phase),
C = partition chromatography (reverse phase),
D = adsorption chromatography,
E = affinity chromatography
F = anion exchange chromatography
G = cation exchange chromatography
What type of liquid chromatography would be best to separate compounds in each of the following categories? **Write the letter** of only one chromatography type (from above) for each of the mixtures below and **give a reason** for your selection.

_____**(i)** Water-insoluble isomers that are less that 2000 molecular mass and have the following general structures (figures to the right):
Reason:



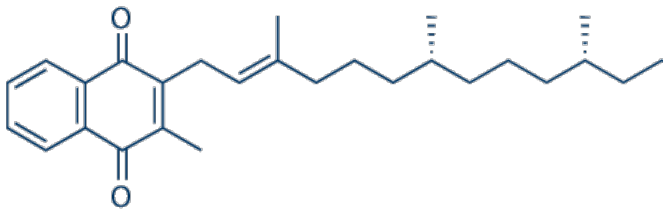
4,Endrin



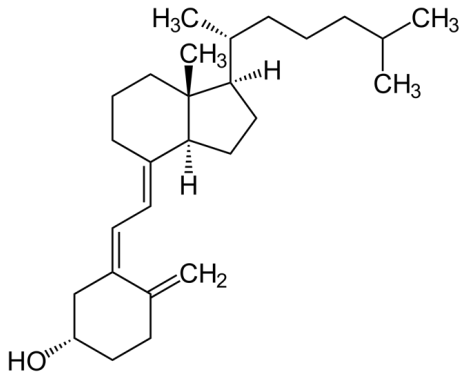
5,Dieldrin

_____**(ii)** Fat-soluble vitamins of similar structures (see examples below), but which are not isomers of each other.

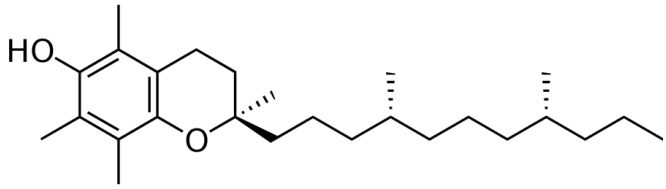
Reason:



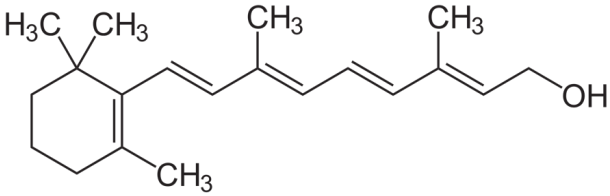
Phylloquinone (Vitamin K_1)



Cholecalciferol (Vitamin D_3)



Alpha tocopherol (Vitamin E)

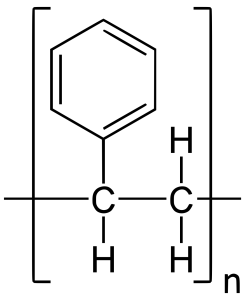


Retinol (Vitamin A)

_____ **(iii)** Different water-soluble anionic molecules from each other in a water solution. There is no desire to separate the different cationic molecules from each other. Where all chemical species are less than 2000 molecular weight.

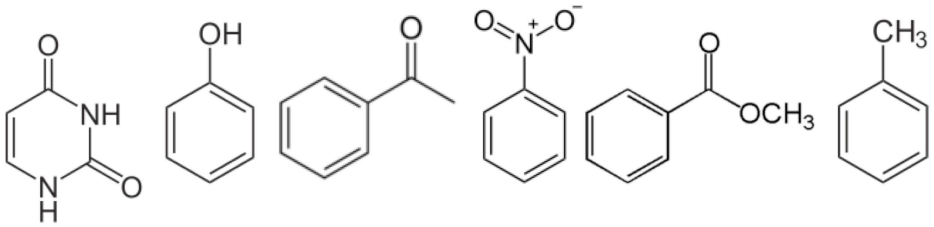
Reason:

_____**(iv)** A sample of synthesized polystyrene molecules of varying lengths (n goes from 7 to 600). (Note that polystyrene is in the material of some coffee cups, take-out containers, and styrofoam packing material). (You are not trying to separate polystyrene based on whether it is syndiotactic or atactic. Just assume that this polystyrene is *atactic*, in which the phenyl groups are randomly distributed on both sides of the polymer chain. Random positioning prevents the chains from aligning with sufficient regularity to achieve crystallinity).



Reason:

_____**(v)** The following series of molecules separated in this order (first on the left, last on the right).



Uracil, phenol, acetophenone, nitrobenzene, methyl benzoate, toluene

Reason:

3. (20 points) A polystyrene resin molecular exclusion HPLC column has a diameter of 6.8 mm and a length of 30 cm. The solid portion of the particles occupies 4% of the volume, the pores occupy 48%, and the volume between particles occupies 48%. Assume that this is an ideal size exclusion column (there is no adsorption).

You separate a mixture of molecules A, B, C, and D using this column. "A" comes off the column at virtually the total exclusion volume of around 5.7 mL. B comes off at 7.8 mL, C comes off at 10.0 mL, and D comes off at about 10.5 mL.

(A) What can you say about the relative molecular mass of B and C? (select one and provide an explanation)

(a) B must be larger than C
(b) C must be larger than B
(c) relative molecular masses of B and C cannot be determined
(d) C and B are the same molecular mass

Explain your selection:

(B) What can you say about the molecular mass of D, if anything? Be quantitative. (select one response and then **show any calculations** to confirm your response)

(a) D is too large for a molecular mass determination
(b) D is too small for a molecular mass determination, it comes off at the total permeation volume
(c) D is larger than B
(d) D is larger than C

Explain your selection and show calculations:

4. (8 points) **<u>Comparison of UPLC with HPLC</u>.** ***Fill in the blanks.*** For each of the following characteristics, write which chromatographic technique it best describes: **UPLC *or* HPLC.** (No explanations are necessary.)

(A) Lower pressures (7,000 psi vs. 19,000 psi)_____

(B) higher sensitivity because peaks are narrower_____

(C) smaller column particles for the stationary phase (1.7 µm vs. 3 µm)_____

(D) Faster run-times (10 min. vs. 30 min.)_____

5. (12 points) Which of the following detectors would you use when performing HPLC using an isocratic elution of a completely unknown sample mixture to get an initial idea of how many major components are in the mixture? This detector responds to nearly all solutes, is considered a bulk property detector, and is inexpensive. Select only one response and explain your selection.

(a) electrochemical detector

(b) fluorescence detector

(c) UV-Vis detector

(d) mass spectrometry detector

(e) differential refractive index detector

(f) flame ionization detector

Explain your response:

6. (20 points) Mass spectrometry is an extremely versatile detection system for GC. However, interfacing an HPLC system to a mass spectrometer is much more challenging. **Describe three major reasons** why it is more difficult to combine HPLC with mass spectrometry than it is to combine GC with mass spectrometry and explain **how those challenges are addressed** to make HPLC-MS feasible. Please **draw a diagram** showing a common interface used for HPLC-MS and explain how it works.